

2024, Mar. 2024 Release LOC:Acute Adult

Withdrawal Syndrome

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Select Day

Initial review ^(1, 2, 3)Episode Day 1 ^(2, 3)Episode Day 2-3 ^(2, 3)Episode Day 4 ^(3, 81)

Episode Day 5

Discharge Screens ⁽⁸³⁾

Utilization Benchmarks

Notes

InterQual® criteria (IQ) is confidential and proprietary information and is being provided to you solely as it pertains to the information requested. IQ may contain advanced clinical knowledge which we recommend you discuss with your physician upon disclosure to you. Use permitted by and subject to license with Change Healthcare LLC and/or one of its subsidiaries. IQ reflects clinical interpretations and analyses and cannot alone either (a) resolve medical ambiguities of particular situations; or (b) provide the sole basis for definitive decisions. IQ is intended solely for use as screening guidelines with respect to medical appropriateness of healthcare services. All ultimate care decisions are strictly and solely the obligation and responsibility of your health care provider. © 2023 Change Healthcare LLC and/or one of its subsidiaries. All Rights Reserved.

Overview

Withdrawal syndrome may occur in patients with a physiological dependence on a substance who abruptly discontinue or decrease their intake. Early signs and symptoms of alcohol withdrawal syndrome include tremors, nightmares, hallucinations, tachycardia, hypertension, arrhythmia, nausea, vomiting, and confusion with agitation. Symptoms can begin as early as 6-8 hours after a decrease in alcohol concentration in the blood, but typically gain severity at 24-36 hours. Delirium tremens is a life-threatening medical emergency that will occur in 5%(.05) of patients experiencing alcohol withdrawal syndrome. Delirium tremens is characterized by hallucinations, impaired consciousness, severe autonomic hypersensitivity, combativeness, hyperthermia, tachycardia, hypertension or hypotension, and seizures. Sedative, hypnotic, and anxiolytic withdrawal is characterized by tachycardia, hypertension, fever, agitation, anxiety, hallucinations, sensory disturbances, nausea, diarrhea, seizures, and delirium.

Instruction: This subset is for patients experiencing an alcohol, anxiolytic, hypnotic, or sedative withdrawal and need acute medical detoxification services. For patients who accidentally or intentionally ingested these substances to a toxic level, see the Toxic Exposure or Ingestion criteria in the General Medical subset. For stabilized patients needing medical and psychiatric services, or patients requiring services related to opioid withdrawal, see the Substance Use Disorders criteria in the Behavioral Health product.

Management of withdrawal syndrome often utilizes protocol-based medication administration to manage the symptoms and side effects associated with withdrawal. In patients with alcohol related withdrawal syndrome, medication administration includes benzodiazepines, beta-blockers, or alpha-2 agonists and is based on the intensity of symptoms. In patients who experience delirium tremens, intubation for airway protection may be necessary. Close management of fluid balance, electrolyte, and metabolic derangements should also be considered. In patients experiencing sedative, hypnotic or anxiolytic withdrawal, management typically includes a substitution and taper protocol to minimize the effects of withdrawal, as well as symptom management. Factors such as underlying anxiety and other psychiatric comorbidities should be anticipated when managing treatment of withdrawal syndrome. If left untreated, reemergence of these symptoms may increase the risk of relapse (McCance-Katz, Psychiatr Serv 2018, 69: 1046-8).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews,

the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2, 3)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **ACUTE, One:** ⁽⁴⁾● Alcohol or anxiolytic or hypnotic or sedative withdrawal syndrome and, **One:** ^(5, 6, 7)● Finding, ≥ **One:**● Alcohol withdrawal syndrome and, ≥ **One:** ⁽⁷⁾– CIWA-Ar 8-14 and blood pressure ≥ 150/90 mmHg ^(8, 9)● Clinical risk factor or finding, ≥ **One:** ⁽¹⁰⁾

– Age > 65 years

– AST/ALT > 1.5 x ULN and > baseline

– Bilirubin > 3 mg/dL

– Chronic kidney disease (includes chronic dialysis) and Creatinine ≥ 2.75 mg/dL

– COPD, Grade 3 or 4 ⁽¹¹⁾– Heart failure, NYHA Class III or Class IV ^(12, 13)– History of seizures or epilepsy ⁽¹⁴⁾– Platelet count < 150,000/cu.mm(150x10⁹/L)

– Positive blood alcohol concentration (BAC)

● Severe and persistent mental illness, cognitive impairment, or substance use disorder (2 or more substances) and, **All:** ^(15, 16, 17, 18)

– Patient unable to complete withdrawal in outpatient setting

– Patient unable to comply with medication-assisted withdrawal management in outpatient setting due to cognitive impairment

– Patient unable to tolerate or manage severity of symptoms in outpatient setting

– Patient unreliable and unable to adhere to outpatient instructions or treatment ⁽¹⁹⁾

– Caregiver unavailable or unable to manage care

● Blood pressure, ≥ **One:**

– Diastolic ≥ 100 mmHg

– Orthostatic systolic drop ≥ 20 mmHg

– Systolic > 150 mmHg ⁽²⁰⁾● Dehydration and, ≥ **One:**

– Blood urea nitrogen (BUN) 25-45 mg/dL(8.9-16.1 mmol/L)

– Creatinine 1.5-3.0 mg/dL(132.6-265.2 μmol/L)

– Heart rate, sustained > 100/min ⁽²¹⁾– Mental status change (excludes coma, stupor, obtundation) or GCS 9-14 ⁽²²⁾● Orthostatic hypotension, **Both:**

– Sustained drop in blood pressure within 3 min of sitting or standing

– Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg

– Sodium > 150 mEq/L(150 mmol/L)

– Urine specific gravity > 1.030

– Vomiting, persistent and unresponsive to ≥ 2 doses antiemetic ⁽²³⁾– Disorientation ^(24, 25)

– Hallucinations

– Heart rate > 100/min, sustained ^(21, 26)– History of complicated withdrawal ⁽²⁷⁾– Myoclonic contractions or tremors ⁽²⁸⁾– Pregnancy and active medical management of withdrawal ⁽²⁹⁾– Seizure ^(30, 31)– Temperature ≥ 101°F(38.3°C) ⁽³²⁾– Vomiting, protracted ⁽³³⁾● **INTERMEDIATE, ≥ One:** ⁽³⁴⁾

- Alcohol withdrawal syndrome and CIWA-Ar 15-20 ^(8, 9, 35, 36)
- Alcohol or anxiolytic or hypnotic or sedative withdrawal syndrome and, ≥ **One:** ^(5, 6, 7, 37)
 - HFNC and respiratory monitoring every 3-4h ^(38, 39)
 - NIPPV ⁽⁴⁰⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽⁴¹⁾
- **CRITICAL, ≥ One:** ⁽⁴²⁾
 - Alcohol withdrawal syndrome and CIWA-Ar > 20 or delirium tremens ^(8, 9, 35, 36)
 - Alcohol or anxiolytic or hypnotic or sedative withdrawal syndrome and, ≥ **One:** ^(5, 6, 7, 37)
 - Anticonvulsant and, ≥ **One:**
 - Recurrent seizure ⁽⁴³⁾
 - Refractory seizure ⁽⁴³⁾
 - Dexmedetomidine, continuous ^(44, 45)
 - HFNC and respiratory monitoring every 1-2h ^(38, 39, 41)
 - Mechanical ventilation
 - NIPPV and, ≥ **One:** ⁽⁴⁰⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
 - Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, ≥ **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 1, One:** ^(2, 3)
(Symptom or finding within 24h)
- **ACUTE, One:** ⁽⁴⁾
 - Alcohol or anxiolytic or hypnotic or sedative withdrawal syndrome and, **Both:** ^(5, 6, 7)
 - Finding, ≥ **One:**
 - Alcohol withdrawal syndrome and, ≥ **One:** ⁽⁷⁾
 - CIWA-Ar 8-14 and blood pressure ≥ 150/90 mmHg ^(8, 9)
 - Clinical risk factor or finding, ≥ **One:** ⁽¹⁰⁾
 - Age > 65 years
 - AST/ALT > 1.5 x ULN and > baseline
 - Bilirubin > 3 mg/dL
 - Chronic kidney disease (includes chronic dialysis) and Creatinine ≥ 2.75 mg/dL
 - COPD, Grade 3 or 4 ⁽¹¹⁾
 - Heart failure, NYHA Class III or Class IV ^(12, 13)
 - History of seizures or epilepsy ⁽¹⁴⁾
 - Platelet count < 150,000/cu.mm(150x10⁹/L)
 - Positive blood alcohol concentration (BAC)
 - Severe and persistent mental illness, cognitive impairment, or substance use disorder (2 or more substances) and, **All:** ^(15, 16, 17, 18)
 - Patient unable to complete withdrawal in outpatient setting
 - Patient unable to comply with medication-assisted withdrawal management in outpatient setting due to cognitive impairment
 - Patient unable to tolerate or manage severity of symptoms in outpatient setting
 - Patient unreliable and unable to adhere to outpatient instructions or treatment ⁽¹⁹⁾
 - Caregiver unavailable or unable to manage care
 - Blood pressure, ≥ **One:**
 - Diastolic ≥ 100 mmHg
 - Orthostatic systolic drop ≥ 20 mmHg
 - Systolic > 150 mmHg ⁽²⁰⁾
 - Dehydration and, ≥ **One:**
 - Blood urea nitrogen (BUN) 25-45 mg/dL(8.9-16.1 mmol/L)
 - Creatinine 1.5-3.0 mg/dL(132.6-265.2 μmol/L)
 - Heart rate, sustained > 100/min ⁽²¹⁾

- Mental status change (excludes coma, stupor, obtundation) or GCS 9-14 ⁽²²⁾
- **Orthostatic hypotension, Both:**
 - Sustained drop in blood pressure within 3 min of sitting or standing
 - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
- Sodium > 150 mEq/L(150 mmol/L)
- Urine specific gravity > 1.030
- Vomiting, persistent and unresponsive to ≥ 2 doses antiemetic ⁽²³⁾
- Disorientation ^(24, 25)
- Hallucinations
- Heart rate > 100 /min, sustained ^(21, 26)
- History of complicated withdrawal ⁽²⁷⁾
- Myoclonic contractions or tremors ⁽²⁸⁾
- Pregnancy and active medical management of withdrawal ⁽²⁹⁾
- Seizure ^(30, 31)
- Temperature $\geq 101^{\circ}\text{F}(38.3^{\circ}\text{C})$ ⁽³²⁾
- Vomiting, protracted ⁽³³⁾
- **Intervention, $\geq$ One:** ^(36, 46)
 - Anticonvulsant (includes PO) ⁽⁴⁷⁾
 - Antipsychotic (includes PO) ⁽⁴⁸⁾
 - Barbiturate (includes PO) ⁽⁴⁹⁾
 - Benzodiazepine (includes PO) ⁽⁵⁰⁾
 - Opiate antagonist (includes PO) ^(51, 52)
 - Opioid (includes PO) ⁽⁵³⁾
 - Sympatholytic (includes PO) ⁽⁵⁴⁾
- **INTERMEDIATE, $\geq$ One:** ⁽³⁴⁾
 - Alcohol withdrawal syndrome and CIWA-Ar 15-20 ^(8, 9, 35, 36)
 - Alcohol or anxiolytic or hypnotic or sedative withdrawal syndrome and, $\geq$ **One:** ^(5, 6, 7, 37)
 - HFNC and respiratory monitoring every 3-4h ^(38, 39)
 - NIPPV ⁽⁴⁰⁾
 - Oxygen $\geq 40\%(0.40)$, $\leq 2\text{d}$ ⁽⁴¹⁾
- **CRITICAL, $\geq$ One:** ⁽⁴²⁾
 - Alcohol withdrawal syndrome and CIWA-Ar > 20 or delirium tremens ^(8, 9, 35, 36)
 - Alcohol or anxiolytic or hypnotic or sedative withdrawal syndrome and, $\geq$ **One:** ^(5, 6, 7, 37)
 - Anticonvulsant and, $\geq$ **One:**
 - Recurrent seizure ⁽⁴³⁾
 - Refractory seizure ⁽⁴³⁾
 - Dexmedetomidine, continuous ^(44, 45)
 - HFNC and respiratory monitoring every 1-2h ^(38, 39, 41)
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ⁽⁴⁰⁾
 - Arterial $\text{Po}_2 \leq 39$ mmHg(5.2 kPa)
 - Arterial or venous $\text{Pco}_2 \geq 60$ mmHg(8.0 kPa) and, $\geq$ **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 2-3, One:** ^(2, 3)
 - **ACUTE, One:**
 - **Early responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁵⁵⁾
 - Afebrile
 - **CIWA-Ar score, $\geq$ One:**
 - Not applicable
 - Within normal limits
 - Laboratory values within acceptable limits

- Medication assisted withdrawal management complete
- Vital signs stable or within acceptable limits ⁽⁵⁶⁾
- Withdrawal symptoms resolving or easily managed
- No complication or active comorbidity relevant to this episode of care ⁽⁵⁷⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Pancreatitis (**use Pancreatitis subset**)
 - Withdrawal unresolved, **Both**:
 - Alcohol or anxiolytic or hypnotic or sedative withdrawal syndrome and, ≥ **One**: ^(5, 6, 7)
 - Alcohol withdrawal syndrome and, ≥ **One**: ⁽⁷⁾
 - CIWA-Ar 8-14 and blood pressure ≥ 150/90 mmHg ^(8, 9)
 - Clinical risk factor and continued alcohol withdrawal management ⁽¹⁰⁾
 - Blood pressure, ≥ **One**:
 - Diastolic ≥ 100 mmHg
 - Orthostatic systolic drop ≥ 20 mmHg
 - Systolic > 150 mmHg ⁽²⁰⁾
 - Dehydration and, ≥ **One**:
 - Blood urea nitrogen (BUN) 25-45 mg/dL(8.9-16.1 mmol/L)
 - Creatinine 1.5-3.0 mg/dL(132.6-265.2 μmol/L)
 - Heart rate, sustained > 100/min ⁽²¹⁾
 - Mental status change (excludes coma, stupor, obtundation) or GCS 9-14 ⁽²²⁾
 - Orthostatic hypotension, **Both**:
 - Sustained drop in blood pressure within 3 min of sitting or standing
 - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
 - Sodium > 150 mEq/L(150 mmol/L)
 - Urine specific gravity > 1.030
 - Vomiting, persistent and unresponsive to ≥ 2 doses antiemetic ⁽²³⁾
 - Disorientation ^(24, 25)
 - Hallucinations or delusions and potential to harm self or others ^(58, 59)
 - Heart rate > 100/min, sustained ^(21, 26)
 - History of complicated withdrawal ⁽²⁷⁾
 - Impaired judgment
 - Impulsivity and psychomotor agitation ^(60, 61)
 - Myoclonic contractions or tremors ⁽²⁸⁾
 - Pregnancy requiring monitoring or management during withdrawal
 - Seizure ^(30, 31)
 - Temperature ≥ 101°F(38.3°C) ⁽³²⁾
 - Vomiting, protracted ⁽³³⁾
 - Intervention, ≥ **One**: ^(36, 46)
 - Anticonvulsant (includes PO) ⁽⁴⁷⁾
 - Antipsychotic (includes PO) ⁽⁴⁸⁾
 - Barbiturate (includes PO) ⁽⁴⁹⁾
 - Benzodiazepine (includes PO) ⁽⁵⁰⁾
 - Opiate antagonist (includes PO) ⁽⁵¹⁾
 - Opioid (includes PO) ⁽⁵³⁾
 - Sympatholytic (includes PO) ⁽⁵⁴⁾
 - Blood glucose ≤ 24h, **All**: ⁽⁶²⁾
 - Finding, ≥ **One**:
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(62, 63)
 - Hepatic encephalopathy Grade II and, **Both**: ^(64, 65, 66)
 - Finding, ≥ **One**:

- Disorientation to time ⁽⁶⁷⁾
- Perceived change in behavior or personality
- Lethargy
- Medication, ≥ **One**:
 - Branched-chain amino acids (BCAA) (includes PO) ⁽⁶⁸⁾
 - Lactulose (includes PO or PR) ⁽⁶⁹⁾
 - L-ornithine L-aspartate (LOLA) (includes PO) ⁽⁷⁰⁾
 - Neomycin (includes PO) ⁽⁷¹⁾
 - Rifaximin (includes PO) ⁽⁷²⁾
- Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One**:
 - Pancreatitis (**use Pancreatitis subset**)
 - Alcohol withdrawal syndrome and CIWA-Ar 15-20 ^(8, 9, 35, 36)
 - Alcohol or anxiolytic or hypnotic or sedative withdrawal syndrome and, ≥ **One**: ⁽⁶⁾
 - HFNC and respiratory monitoring every 3-4h ^(38, 39)
 - NIPPV ⁽⁴⁰⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽⁴¹⁾
 - Arrhythmia, ≥ **One**:
 - IV antiarrhythmic and hemodynamically stable
 - Antiarrhythmic conversion from IV to PO, **One**:
 - No proarrhythmic risk ≤ 24h
 - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d ⁽⁷³⁾
 - Hepatic encephalopathy Grade III and, **Both**: ^(64, 65)
 - Finding, ≥ **One**:
 - Unusual or bizarre behavior ⁽⁷⁴⁾
 - Confused
 - Gross disorientation ⁽⁷⁵⁾
 - Responsive to noxious stimuli only
 - Somnolence
 - Medication, ≥ **One**: ⁽⁷⁶⁾
 - Branched-chain amino acids (BCAA) (includes PO) ⁽⁶⁸⁾
 - Lactulose (includes PO or PR) ⁽⁶⁹⁾
 - L-ornithine L-aspartate (LOLA) ⁽⁷⁰⁾
 - Neomycin (includes PO) ⁽⁷¹⁾
 - Rifaximin (includes PO) ⁽⁷²⁾
- **CRITICAL, ≥ One**:
 - Pancreatitis (**use Pancreatitis subset**)
 - Alcohol withdrawal syndrome and CIWA-Ar > 20 or delirium tremens ^(8, 9, 35, 36)
 - Anticonvulsant and, ≥ **One**:
 - Recurrent seizure ⁽⁴³⁾
 - Refractory seizure ⁽⁴³⁾
 - Arrhythmia, ≥ **One**:
 - Urgent electrical cardioversion performed within last 24h
 - Defibrillation performed within last 24h
 - Temporary pacemaker insertion performed within last 2d
 - Symptomatic recurrent or refractory arrhythmia requiring antiarrhythmic and, ≥ **One**: ⁽⁷⁷⁾
 - Chest pain
 - Heart failure and, ≥ **One**:
 - Arterial Po₂ < 56 mmHg(7.4 kPa)
 - O₂ sat < 89%(0.89) and < baseline
 - Hemodynamic instability, ≥ **One**:
 - Heart rate > 120/min, sustained ⁽²¹⁾
 - Systolic blood pressure, ≥ **One**:

- < 90 mmHg
- < 110 mmHg with chronic hypertension
- > 30 mmHg decrease from baseline ⁽⁷⁸⁾
- Labile ⁽⁷⁹⁾
- MAP < 65 mmHg
- Dexmedetomidine, continuous ^(44, 45)
- Hepatic encephalopathy Grade IV and coma and, ≥ **One:** ^(64, 65, 76)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽⁶⁸⁾
 - Lactulose (includes PO or PR) ⁽⁶⁹⁾
 - L-ornithine L-aspartate (LOLA) ⁽⁷⁰⁾
 - Neomycin (includes PO) ⁽⁷¹⁾
 - Rifaximin (includes PO) ⁽⁷²⁾
- HFNC and respiratory monitoring every 1-2h ^(38, 39, 41)
- Mechanical ventilation
- NIPPV and, ≥ **One:** ⁽⁴⁰⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁸⁰⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 4, One:** ^(3, 81)
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁵⁵⁾
 - Afebrile
 - **CIWA-Ar score, ≥ One:**
 - Not applicable
 - Within normal limits
 - Laboratory values within acceptable limits
 - Medication assisted withdrawal management complete
 - Vital signs within acceptable limits ⁽⁵⁶⁾
 - Withdrawal symptoms resolving or easily managed
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁵⁷⁾
 - **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - Blood glucose within acceptable limits ⁽⁵⁶⁾
 - **Hepatic encephalopathy and, Both:** ⁽⁶⁴⁾
 - Grade I or within acceptable limits ⁽⁵⁶⁾
 - Neurological stability last 24h ⁽⁸²⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Pancreatitis (**use Pancreatitis subset**)
 - **Withdrawal unresolved, Both:**
 - **Alcohol or anxiolytic or hypnotic or sedative withdrawal syndrome and, ≥ One:** ^(5, 6, 7)
 - **Alcohol withdrawal syndrome and, ≥ One:** ⁽⁷⁾
 - CIWA-Ar 8-14 and blood pressure ≥ 150/90 mmHg ^(8, 9)
 - Clinical risk factor and continued alcohol withdrawal management ⁽¹⁰⁾
 - **Blood pressure, ≥ One:**
 - Diastolic ≥ 100 mmHg
 - Orthostatic systolic drop ≥ 20 mmHg
 - Systolic > 150 mmHg ⁽²⁰⁾
 - **Dehydration and, ≥ One:**
 - Blood urea nitrogen (BUN) 25-45 mg/dL(8.9-16.1 mmol/L)
 - Creatinine 1.5-3.0 mg/dL(132.6-265.2 μmol/L)
 - Heart rate, sustained > 100/min ⁽²¹⁾
 - Mental status change (excludes coma, stupor, obtundation) or GCS 9-14 ⁽²²⁾
 - **Orthostatic hypotension, Both:**

- Sustained drop in blood pressure within 3 min of sitting or standing
- Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
- Sodium > 150 mEq/L(150 mmol/L)
- Urine specific gravity > 1.030
- Vomiting, persistent and unresponsive to ≥ 2 doses antiemetic ⁽²³⁾
- Disorientation ^(24, 25)
- Hallucinations or delusions and potential to harm self or others ^(58, 59)
- Heart rate > 100 /min, sustained ^(21, 26)
- History of complicated withdrawal ⁽²⁷⁾
- Impaired judgment
- Impulsivity and psychomotor agitation ^(60, 61)
- Myoclonic contractions or tremors ⁽²⁸⁾
- Pregnancy requiring monitoring or management during withdrawal
- Seizure ^(30, 31)
- Temperature $\geq 101^\circ\text{F}(38.3^\circ\text{C})$ ⁽³²⁾
- Vomiting, protracted ⁽³³⁾
- Intervention, $\geq$ **One:** ^(36, 46)
 - Anticonvulsant (includes PO) ⁽⁴⁷⁾
 - Antipsychotic (includes PO) ⁽⁴⁸⁾
 - Barbiturate (includes PO) ⁽⁴⁹⁾
 - Benzodiazepine (includes PO) ⁽⁵⁰⁾
 - Opiate antagonist (includes PO) ⁽⁵¹⁾
 - Opioid (includes PO) ⁽⁵³⁾
 - Sympatholytic (includes PO) ⁽⁵⁴⁾
- Blood glucose $\leq 24\text{h}$, **All:** ⁽⁶²⁾
 - Finding, $\geq$ **One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(62, 63)
- Hepatic encephalopathy Grade II and, **Both:** ^(64, 65, 66)
 - Finding, $\geq$ **One:**
 - Disorientation to time ⁽⁶⁷⁾
 - Perceived change in behavior or personality
 - Lethargy
 - Medication, $\geq$ **One:**
 - Branched-chain amino acids (BCAA) (includes PO) ⁽⁶⁸⁾
 - Lactulose (includes PO or PR) ⁽⁶⁹⁾
 - L-ornithine L-aspartate (LOLA) (includes PO) ⁽⁷⁰⁾
 - Neomycin (includes PO) ⁽⁷¹⁾
 - Rifaximin (includes PO) ⁽⁷²⁾
- Post Critical care $\leq 24\text{h}$
- INTERMEDIATE, $\geq$ **One:**
 - Pancreatitis (**use Pancreatitis subset**)
 - Alcohol withdrawal syndrome and CIWA-Ar 15-20 ^(8, 9, 35, 36)
 - Alcohol or anxiolytic or hypnotic or sedative withdrawal syndrome and, $\geq$ **One:** ⁽⁶⁾
 - HFNC and respiratory monitoring every 3-4h ^(38, 39)
 - NIPPV ⁽⁴⁰⁾
 - Oxygen $\geq 40\%(0.40)$, $\leq 2\text{d}$ ⁽⁴¹⁾
 - Arrhythmia, $\geq$ **One:**
 - IV antiarrhythmic and hemodynamically stable
 - Antiarrhythmic conversion from IV to PO, **One:**
 - No proarrhythmic risk $\leq 24\text{h}$

- Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) $\leq 3d$ ⁽⁷³⁾
- Hepatic encephalopathy Grade III and, **Both:** ^(64, 65)
 - Finding, $\geq$ **One:**
 - Unusual or bizarre behavior ⁽⁷⁴⁾
 - Confused
 - Gross disorientation ⁽⁷⁵⁾
 - Responsive to noxious stimuli only
 - Somnolence
 - Medication, $\geq$ **One:** ⁽⁷⁶⁾
 - Branched-chain amino acids (BCAA) (includes PO) ⁽⁶⁸⁾
 - Lactulose (includes PO or PR) ⁽⁶⁹⁾
 - L-ornithine L-aspartate (LOLA) ⁽⁷⁰⁾
 - Neomycin (includes PO) ⁽⁷¹⁾
 - Rifaximin (includes PO) ⁽⁷²⁾
- **CRITICAL, $\geq$ One:**
 - Pancreatitis (**use Pancreatitis subset**)
 - Alcohol withdrawal syndrome and CIWA-Ar > 20 or delirium tremens ^(8, 9, 35, 36)
 - Anticonvulsant and, $\geq$ **One:**
 - Recurrent seizure ⁽⁴³⁾
 - Refractory seizure ⁽⁴³⁾
 - Arrhythmia, $\geq$ **One:**
 - Urgent electrical cardioversion performed within last 24h
 - Defibrillation performed within last 24h
 - Temporary pacemaker insertion performed within last 2d
 - Symptomatic recurrent or refractory arrhythmia requiring antiarrhythmic and, $\geq$ **One:** ⁽⁷⁷⁾
 - Chest pain
 - Heart failure and, $\geq$ **One:**
 - Arterial $PO_2 < 56$ mmHg(7.4 kPa)
 - O_2 sat $< 89\%$ (0.89) and $<$ baseline
 - Hemodynamic instability, $\geq$ **One:**
 - Heart rate > 120 /min, sustained ⁽²¹⁾
 - Systolic blood pressure, $\geq$ **One:**
 - < 90 mmHg
 - < 110 mmHg with chronic hypertension
 - > 30 mmHg decrease from baseline ⁽⁷⁸⁾
 - Labile ⁽⁷⁹⁾
 - MAP < 65 mmHg
 - Dexmedetomidine, continuous ^(44, 45)
 - Hepatic encephalopathy Grade IV and coma and, $\geq$ **One:** ^(64, 65, 76)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽⁶⁸⁾
 - Lactulose (includes PO or PR) ⁽⁶⁹⁾
 - L-ornithine L-aspartate (LOLA) ⁽⁷⁰⁾
 - Neomycin (includes PO) ⁽⁷¹⁾
 - Rifaximin (includes PO) ⁽⁷²⁾
 - HFNC and respiratory monitoring every 1-2h ^(38, 39, 41)
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ⁽⁴⁰⁾
 - $FiO_2 > 50\%$ (0.50)
 - Respiratory intervention every 1-2h ⁽⁸⁰⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 5, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁵⁵⁾

- Afebrile
- CIWA-Ar score, **≥ One**:
 - Not applicable
 - Within normal limits
- Laboratory values within acceptable limits
- Medication assisted withdrawal management complete
- Vital signs within acceptable limits ⁽⁵⁶⁾
- Withdrawal symptoms resolving or easily managed
- Complication or comorbidity, **One**:
 - No complication or active comorbidity relevant to this episode of care ⁽⁵⁷⁾
 - Complication or comorbidity and clinically stable for discharge, **≥ One**:
 - Blood glucose within acceptable limits ⁽⁵⁶⁾
 - Hepatic encephalopathy and, **Both**: ⁽⁶⁴⁾
 - Grade I or within acceptable limits ⁽⁵⁶⁾
 - Neurological stability last 24h ⁽⁸²⁾
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One**:
 - Use Extended Stay subset
 - Refer for secondary review
- **INTERMEDIATE, One**:
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One**:
 - Use Extended Stay subset
 - Refer for secondary review
- **CRITICAL, One**:
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One**:
 - Use Extended Stay subset
 - Refer for secondary review

Discharge, Level of Care, One: ⁽⁸³⁾

- **HOME, Both**:
 - Level of care appropriateness, **Both**:
 - Home environment safe and accessible ⁽⁸⁴⁾
 - Patient or caregiver demonstrates ability to manage care
 - Withdrawal syndrome discharge planning, **All**:
 - Provide comprehensive written discharge and teaching instructions ⁽⁸⁵⁾
 - Medication reconciliation ⁽⁸⁶⁾
 - Follow-up care planned
 - Identify and address transportation needs
- **HOME CARE, Both**:
 - Level of care appropriateness, **All**:
 - Home environment safe and accessible ⁽⁸⁴⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(87, 88)
 - Skilled services, **≥ One**: ⁽⁸⁹⁾
 - Skilled nursing ⁽⁹⁰⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁹¹⁾
 - Withdrawal syndrome discharge planning, **All**:
 - Provide comprehensive written discharge and teaching instructions ⁽⁸⁵⁾
 - Medication reconciliation ⁽⁸⁶⁾
 - Follow-up care planned

- Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽⁹²⁾
 - Skilled treatment, **≥ One:**
 - Clinical assessment ⁽⁹³⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁹⁴⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 1\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment $1\text{-}2\text{x}/24\text{h}$ ⁽⁹⁰⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(95, 96)
 - Functional impairments requiring at least supervision ⁽⁹⁷⁾
 - Able to tolerate 1h/d of skilled therapy $\geq 5\text{d/wk}$
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 2\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment $4\text{h}/24\text{h}$ ⁽⁹⁰⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(95, 96)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁹⁷⁾
 - Able to tolerate $2\text{-}3\text{h/d}$ of skilled therapy $\geq 5\text{d/wk}$ and ≥ 1 therapy discipline
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(95, 96)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁹⁷⁾
 - Able to tolerate $2\text{-}3\text{h/d}$ of skilled therapy $\geq 5\text{d/wk}$ and ≥ 2 therapy disciplines
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - Level of care appropriateness, **Both:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
- **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(95, 96)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁹⁷⁾
 - Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽⁹⁸⁾
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁸⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - **Level of care appropriateness, Both:**
 - Treatment precluded in a lower level
 - **In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁹⁹⁾
 - Ventilator dependent $\geq 6h/d$ and weaning planned ^(100, 101)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽¹⁰²⁾
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁸⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
894 ALCOHOL, DRUG ABUSE OR DEPENDENCE, LEFT AMA	n/a	2.1	CMS GMLOS
895 ALCOHOL, DRUG ABUSE OR DEPENDENCE WITH REHABILITATION THERAPY	n/a	8.3	CMS GMLOS
896 ALCOHOL, DRUG ABUSE OR DEPENDENCE WITHOUT REHABILITATION THERAPY WITH MCC	n/a	5	CMS GMLOS
897 ALCOHOL, DRUG ABUSE OR DEPENDENCE WITHOUT REHABILITATION THERAPY WITHOUT MCC	n/a	3.4	CMS GMLOS
ALCOHOL WITHDRAWAL SYNDROME	9%	4.2	InterQual
SEDATIVE, HYPNOTIC OR ANXIOLYTIC WITHDRAWAL SYNDROME	15%	4.3	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

2:**Expected progress after 24-48h**

- Vital signs normalizing and within acceptable limits
- Urgent behavioral health issue addressed and returning to acceptable limits (e.g., unstable psychiatric condition that places the patient at risk of injury to himself or others)
- Fluid, electrolyte imbalances, and nutrition returning to acceptable limits
- Labs returning to acceptable limits
- PO intake improving
- Activity increasing
- Medication regimen stabilized and tolerated
- Preparing for discharge

Consider

- Education related to abstinence planning and relapse prevention
- Behavioral Health assessment
- Change route of medication to PO
- Discharge planning

3:**Care Facilitation**

- Home nutrition therapy should be considered in a clinically stable patient who continues to require supplemental nutrition due to the inability to tolerate a nutritious, solid diet (**Home Care**)
- Referral to community-based recovery groups or activities (e.g., Alcoholics Anonymous, Narcotics Anonymous)
- Referral to rehabilitation program (e.g., community-based recovery groups or activities, residential treatment center, supervised living, partial hospital program, intensive outpatient program, outpatient program)

Potential risk for admission, consider an alternate level of care or readmission program.

- Knowledge deficit or nonadherence with condition management identified (**Home Care, Readmission Reduction Program**). Include patient education focusing on medication regimen, nutritional teaching, psychosocial stressors, and evaluation for substance use disorder.
- Patient who has an inability to tolerate a solid diet, gastrointestinal symptoms (e.g., nausea, vomiting, diarrhea) (**Home Care, Readmission Reduction Program**)
- Hospital admissions within the last 6 months (**Home Care, Readmission Reduction Team**)
- Comorbid diabetes, pancreatitis, infection, cirrhosis, gastrointestinal bleeding, nutritional deficiencies (**Home Care, Readmission Reduction Team**)

Social Determinants of Health (SDOH)

Social Determinants of Health (SDOH) are conditions in a person's environment that may affect their health. Social needs related to SDOH may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. Children and caregivers should be screened for SDOH-related barriers to health to mitigate any unmet social and psychosocial factors impacting the health of the child and family unit (Kreuter et al., Annu Rev Public Health 2021, 42: 329-44; Sokol et al., Pediatrics 2019, 144:). The following social needs should be assessed on admission to determine which interventions are relevant to the individual child and their family support system. This list may not be all inclusive and all relevant SDOH factors should be considered. Any social need determined to be relevant to a given child and family should be planned for throughout the hospital stay.

*Interventions should be considered for all patients and families if social need is relevant

Alcohol or substance use

- Coordinate care with outpatient primary care or behavioral health providers*
- Refer to addiction social support systems*
- Provide information about community resources and programs for family-centered treatment (e.g., intensive case management, detox, outpatient, residential treatment, day treatment, and medication assisted treatment)
- Provide information about home visitation programs (e.g., Parent-Child Assistance Program (PCAP) for pregnant and parenting individuals with substance use disorders aftercare) (Hildebrandt et al., Infant Ment Health J 2020, 41: 677-96; Vanderplasschen et al., Front Psychiatry 2019, 10: 186)
- Provide information about Cognitive Behavioral Therapy (CBT) Multidimensional Family Therapy (MDFT) and/or Pharmacotherapy for youth with co-occurring emotional/mental disorders (Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021)

Child maltreatment, abuse, or intimate partner abuse - sexual and/or psychological abuse

- Arrange for alternative housing situation, foster care, or shelter if home environment is unsafe*
- Assist with creating a long-term safety plan or make appropriate referral*
- Refer to child protective services (CPS) and other domestic violence services if warranted*
- Refer for psychosocial support if warranted*
- Provide community resources and legal information
- Provide information on parent training programs such as the Nurse Family Partnership Program (NFP) (Nurse-Family Partnership National Resources for Families. 2021; Gubbels et al., Int J Environ Res Public Health 2019, 16:)

End of life planning, arrange palliative care or hospice

- Consider palliative care for children with end stage disease
- Consider hospice care for terminally ill children with a life expectancy of less than 6 months

Environmental exposure to toxic substances and other physical hazards

Lead poisoning:

- Provide community resources for primary prevention and secondary prevention strategies (e.g., blood lead level testing and follow up (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Tobacco use (e.g., cigarettes, E-cigarettes, vaping):

- Provide education and/or brief counseling about tobacco use cessation and the risk of second-hand smoke (U.S. Department of Health and Human Services, Tobacco Use in Children and Adolescents: Primary Care Interventions. 2021)

Financial resources strained or limited

Food insecurity:

- Provide information about local food banks and special nutritional programs (e.g., Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), U.S. Department of Agriculture Summer Food Service Program (Testa and Jackson, Ann Epidemiol 2021, 58: 22-8; Berman et al., Pediatr Rev 2018, 39: 235-46)
- Refer to direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)

Income insecurity:

- Identify income resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Refer to direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)

Infants and children with Special Health Care Needs

- Provide anticipatory guidance and education to caregivers regarding medical condition and expected progress after discharge over the next few days, months, or years, as warranted (Dosman and Andrews, Paediatr Child Health 2012, 17: 75-80)*
- Refer for follow-up care to neonatologist or other medical subspecialist for infants or children with ongoing and/or chronic medical issues, (e.g., bronchopulmonary dysplasia or feeding dysfunction) (Pediatrics 2008, 122: 1119-26)*
- Refer for adaptive/assistive medical equipment and equipment for technology-dependent children, (e.g., home oxygen, ventilators, monitors, or tube feeding)(Pediatrics 2008, 122: 1119-26)*
- Refer for neurodevelopmental assessment for high-risk infants*
- Refer to post-discharge follow-up programs (e.g., Maternal, Infant, and Early Childhood Home Visiting Program, Early Head Start program for infants or toddlers under the age of 3, and pregnant individuals in need of medical, mental health, nutrition and education, or the Nurse-Family Partnership for mothers pregnant with their first child and for ongoing nurse care visits through the child's second birthday) (Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021; Nurse-Family Partnership National Resources for Families. 2021)
- Refer for behavioral support and/or counseling for caregivers through therapy groups, telephone support, and/or peer support groups for caregivers of children with similar health needs (Reyes et al., Health Equity 2021, 5: 100-18; Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021; Pierron et al., BMC Public Health 2018, 18: 1087)
- Refer to lactation consultant in the hospital and home visits for infants at risk for poor lactation or suboptimal infant breast-feeding behavior (Leeman et al., Pediatr Qual Saf 2019, 4: e130)
- Refer for respite services, if warranted

Mental health co-morbidities involved

- Arrange appointment with primary care provider or behavioral health specialist upon discharge*
- Coordinate care with outpatient behavioral health providers*
- Provide information about suicide prevention, including the National Suicide Prevention hotline*
- Refer for psychiatric services for suicide risk screening and intervention
- Refer to school-based mental health interventions, if applicable (Fazel et al., Lancet Psychiatry 2014, 1: 377-87)
- Provide information about Intensive Community-based treatment (e.g., Functional Family Therapy, Intensive Child and Adolescent and Psychiatric Services, Multidimensional Family Therapy, and Intensive Case Management)
- Transfer to day treatment program, residential addiction services or partial hospitalization when warranted

Risk for treatment nonadherence

- Provide information about educational and behavioral support initiatives for children at risk of medication nonadherence and their caregivers, such as text message reminders to take medication and electronic medication dispensers
- Provide information about peer support groups for children and/or families with poor insight into illness or noncompliance with treatment (Mental Health America, Peer Support: Research and Reports 2021; National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- Refer to interventions to prevent medical relapse, or readmission, and to enhance medication compliance

Risk for readmission

- Consider transfer to an alternative level of care such as subacute care
- Consider home care services
- Consider early childhood home visiting programs (e.g., Early Head Start; Early On; Nurse Family

Partnership; and the Maternal, Infant, and Early Childhood Home Visiting Program (Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021; Nurse-Family Partnership National Resources for Families. 2021)

- Consider telehealth and web-based training programs (e.g., telepsychiatry or telebehavioral health and mobile health interventions (J Am Acad Child Adolesc Psychiatry 2017, 56: 875-93)
- Consider Intensive in-home behavioral health services
- Provide patient and family with condition-specific information including appropriate planning for prevention of RSV infection and appropriate immunizations

Special Health Risk Behaviors in adolescence and young adulthood

- Arrange follow-up appointment with primary care provider and/or specialists upon discharge*
- Coordinate care with outpatient behavioral health providers*
- Provide information on mentoring programs, (e.g., Big Brothers Big Sisters of America)
- Refer for Intensive Community-Based treatment for children with comorbid behavioral health issues
- Refer to community resources for teen pregnancy prevention/family planning (use of birth control, subsequent pregnancies) (Maness et al., J Adolesc Health 2016, 58: 636-43)

For eating disorders:

- Provide information on nutrition services including education and resources for diet management, physical activity, and promotion of healthy eating attitudes and behaviors (Gato-Moreno et al., Int J Environ Res Public Health 2021, 18;; Al-Khudairy et al., Cochrane Database Syst Rev 2017, 6: Cd012691)
- Provide information on outpatient psychosocial interventions and Family-Based Treatment (Lock and La Via, J Am Acad Child Adolesc Psychiatry 2015, 54: 412-25)

For parenting resources:

- Provide information on Family Resource Centers and Parenting support groups for young, first-time parents, (e.g., National Parent Helpline) (National Parent and Youth Helpline)
- Provide information on community resources to improve parenting skills (face-to-face or by telephone) (U.S. Department of Health and Human Services, Parenting Resources. 2021)

Unstable living environment or homelessness expected upon discharge

- Assist with health insurance application
- Provide information about homeless shelter application to caregivers if applicable
- Refer to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- Refer to federally sponsored outpatient programs
- Refer to Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community Treatment (Ponka et al., PLoS One 2020, 15: e0230896; Vijverberg et al., BMC Psychiatry 2017, 17: 284)

4:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

5:

When a physically addicting drug is reduced or stopped abruptly, an individual may develop physiological and psychological withdrawal symptoms. These withdrawal symptoms are usually the opposite of the intoxicating symptoms of the same drug. Significant withdrawal syndromes are seen with alcohol, anxiolytics, hypnotic, opioids and sedatives and synthetic cannabinoids. Withdrawal syndromes differ for each specific kind or classification of drug (Cooper, Curr Psychiatry Rep 2016, 18: 52).

6:

Examples of sedatives/hypnotics/anxiolytics include, but are not limited to:

Barbiturates:

- Amobarbital (Amytal®)

- Butalbital (Fiorinal®)
- Pentobarbital (Nembutal®)
- Phenobarbital
- Secobarbital (Seconal®)

Benzodiazepines:

- Alprazolam (Xanax®)
- Chlordiazepoxide (Librium®)
- Clonazepam (Klonopin®)
- Diazepam (Valium®)
- Flunitrazepam (Rohypnol®)
- Lorazepam (Ativan®)
- Oxazepam (Serax®)

Depressants or anesthetics:

- Gamma-butyrolactone (GBL) - a precursor molecule to GHB, marketed as "verve" or "jolt" especially via the internet
- Gamma-hydroxybutyrate (GHB) - also marketed illegally in the U.S. as a body building aid or sleep aid*#

Depressants or sedative-hypnotics:

- Carisoprodol (Soma®)
- Chloral hydrate
- Eszopiclone (Lunesta®)
- Glutethimide (Doriden®)
- Meprobamate (Equanil® or Miltown®)
- Methaqualone (Quaalude®)
- Zaleplon (Sonata®)
- Zolpidem (Ambien® / Ambien CR®)

*Rohypnol®, Xanax®, Klonopin®, and GHB are currently popular as club drugs used at raves, trances, and circuit parties, and in clubs.

*The club drugs GHB, Rohypnol®, and Klonopin® are sometimes termed "date rape" drugs because when they are mixed with alcohol, the individual may become incapacitated due to the cumulative sedative effects and thus is unable to resist a sexual assault. When these drugs are mixed with alcohol, they may cause amnesia and the individual may not be able to recall events that occurred while intoxicated (Busardo et al., Eur Rev Med Pharmacol Sci 2019, 23: 10577-87).

7:

Patients in alcohol withdrawal can present with varying symptoms including but not limited to:

- Anxiety
- Insomnia
- Nausea/vomiting
- Autonomic hyperactivity
- Agitation
- Tremor
- Seizure
- Hallucinations
- Delirium

Although clinical judgment ultimately determines the classification of withdrawal severity, different patients and patient populations can have an impact on the physicians' reference range (J Addict Med 2020, 14: 1-72; Melkonian et al., Mayo Clin Proc Innov Qual Outcomes 2019, 3: 344-9).

8:

There are various standardized scales measuring the severity of alcohol withdrawal. The withdrawal scores cited in the criteria are based on the Clinical Institute Withdrawal Assessment for Alcohol revised scale (CIWA-Ar) (American Society of Addiction Medicine, The ASAM Clinical Practice Guideline on Alcohol Withdrawal Management. 2020; Grover and Ghosh, J Clin Exp Hepatol 2018, 8: 460-70; Gee et al., International Journal of

Research in Nursing 2017, 8: 17-22). The CIWA-Ar has been analyzed extensively and validated in a number of studies (Melkonian et al., Mayo Clin Proc Innov Qual Outcomes 2019, 3: 344-9; Eloma et al., Am J Drug Alcohol Abuse 2018, 44: 418-25). The CIWA-Ar scale and many of its modified versions assess signs and symptoms of withdrawal, including observed tremor, nausea, and vomiting, sweating, clouding of the sensorium, hallucinations, quality of contact, anxiety, and agitation. This assessment scale can assist in determining the patient's severity of withdrawal and the likelihood of experiencing alcohol withdrawal syndrome, which can lead to delirium tremens or seizures (Wolf et al., Open Access Emerg Med 2020, 12: 53-65; Attilia et al., Riv Psichiatr 2018, 53: 118-22; Grover and Ghosh, J Clin Exp Hepatol 2018, 8: 460-70; Perry Christopher, The Management of Substance Use Disorders: Synopsis of the 2021 U.S Department of Veterans Affairs and U.S. Department of Defense Clinical Practice Guideline. 2022). A numeric value is assigned to each depending on the severity of the signs and symptoms. Other standardized withdrawal scoring instruments are not considered equivalent and cannot be substituted to apply criteria. Medical staff should be adequately and regularly trained on the administration of the CIWA-Ar (VandenBerg et al., Mental Health Clinician 2016, 6: 114-9).

9:

Organizational policy determines the frequency of monitoring patients actively withdrawing. Monitoring frequency varies based on clinical findings, clinician judgment, and patient needs.

10:

Consideration of clinical risk factors, the presence and severity of comorbid illnesses and their probability of complicating alcohol withdrawal management is a component of level of care determination. Inpatient setting management should be utilized for patients with active and potentially unstable medical problems. Older age, elevated liver enzymes and unstable comorbid illnesses such as chronic kidney disease (includes dialysis), COPD Grade 3 or 4, heart failure NYHA Class III or Class IV, epilepsy or generalized seizure, severe and persistent mental illness, cognitive impairment, and substance use disorder (two or more substances) are risk factors for complicated alcohol withdrawal (J Addict Med 2020, 14: 1-72).

11:

The severity of chronic obstructive pulmonary disease (COPD) airflow limitations is classified into 4 grades based on the patient's baseline forced expiratory volume (FEV₁) measurement post-bronchodilator administration as follows (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024):

- GOLD 1 - Mild: FEV₁ ≥ 80%(0.80) predicted
- GOLD 2 - Moderate: FEV₁ 50-79%(0.50-0.79) predicted
- GOLD 3 - Severe: FEV₁ 30-49%(0.30-0.49) predicted
- GOLD 4 - Very severe: FEV₁ < 30%(0.30) predicted

12:

New York Heart Association (NYHA) classification for heart failure is defined as follows (Yap et al., Clinical cardiology 2015, 38: 621-8; New York Heart Association, Diseases of the Heart and Blood Vessels. Nomenclature and Criteria for diagnosis. 1964):

- Class I - No limitation of physical activity. Ordinary physical activity does not cause undue fatigue, palpitation, dyspnea, or anginal pain.
- Class II - Slight limitation of physical activity. Comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnea, or anginal pain.
- Class III - Marked limitation of physical activity. Comfortable at rest. Less than ordinary physical activity causes fatigue, palpitation, dyspnea, or anginal pain.
- Class IV - Inability to carry on any physical activity without discomfort. Symptoms of cardiac insufficiency or of anginal syndrome may be present even at rest. If any physical activity is undertaken, discomfort is increased.

13:

Patients with chronic obstructive pulmonary disease (COPD) and congestive heart failure (CHF) as a comorbid condition have greater morbidity and mortality when hospitalized (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

14:

Patients with history of seizure are at higher risk of experiencing withdrawal seizures (J Addict Med 2020, 14: 1-72).

15:

Patients hospitalized with heart failure, acute myocardial infarction, or pneumonia in addition to a psychiatric comorbidity have higher readmission rates within 30 days than those without a psychiatric diagnosis (Ahmedani et al., Psychiatr Serv 2015, 66: 134-40).

16:

The term "severe and persistent mental illness, cognitive impairment" also called "serious and persistent mental illness," refers to a mental illness or cognitive impairment that is characterized by severe symptoms of prolonged duration that require long-term treatment. The illness causes impaired functioning that interferes significantly with primary activities of daily living (ADLs) and results in an inability to maintain independent functioning without treatment, support, and rehabilitation for a prolonged or indefinite period of time. Examples include, but are not limited to, schizophrenia and other psychotic disorders, bipolar disorder, dementia, Alzheimer's disease, and severe cases of major depression.

17:

Cognitive impairment includes, but is not limited to:

- Difficulty exercising judgment
- Difficulty planning and carrying out tasks
- Difficulty understanding the meaning or importance of things that they normally understand
- Impaired ability to learn new things
- Impaired memory
- Impaired thinking skills
- Not recognizing familiar people and places

18:

Substance use, including but not limited to marijuana, stimulants, alcohol, hallucinogens, and substance use (two or more substances) may cause serious cognitive impairment and nonadherence to treatment (Rolland et al., Neuropsychol Rev 2019, 29: 103-15). Cognitive impairment may also negatively impact target areas of treatment such as readiness for change, self-efficacy, insight, and coping skills application (Rolland et al., Neuropsychol Rev 2019, 29: 103-15).

19:

Patient unreliable refers to the following:

- Inability or unwillingness to follow an outpatient monitoring or treatment plan
- Inability to understand the signs of deterioration and recognize the need for prompt medical attention

20:

Individuals with alcohol withdrawal syndrome (AWS) and a systolic blood pressure greater than 150 may be at increased risk of developing delirium tremens (Day and Daly, Addiction 2022, 117: 804-14; Wood et al., JAMA 2018, 320: 825-33).

21:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with

diverse expertise in varied geographic settings.

22:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

23:

Patients with mild to moderate active vomiting should be treated with intravenous hydration and antiemetic therapy. A dehydrated patient who cannot tolerate oral hydration should not be discharged (Joshua Guttman, In: Rosen's Emergency Medicine, 9 edn. 2018:230-41.e1).

24:

Disorientation refers to confusion about who the person is, place, and time (American Psychiatric Association, Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition Text Revision, DSM-5-TR, 2022).

25:

Patients experiencing alcohol withdrawal who are found to be disoriented require close monitoring. Disorientation and agitation can worsen and require more frequent doses of benzodiazepines (J Addict Med 2020, 14: 1-72).

26:

Individuals with alcohol withdrawal syndrome and a heart rate greater than 100 beats per minute may be at increased risk of developing delirium tremens (Lee et al., J Gastroenterol Hepatol 2005, 20: 1833-7).

27:

Complicated withdrawal is a term used to describe a withdrawal episode which resulted in severe/life-threatening symptoms such as withdrawal-related seizures or delirium. Patients with a known history of having experienced a complicated withdrawal are at high risk of complicated withdrawal recurrence and should be admitted to the acute level of care (J Addict Med 2020, 14: 1-72).

28:

Myoclonic contractions are characterized by clonic spasms or twitching of a muscle(s) and can be a symptom of severe alcohol withdrawal (Day and Daly, Addiction 2022, 117: 804-14; Wood et al., JAMA 2018, 320: 825-33).

29:

Patients may require treatment on a medical unit rather than a psychiatric or substance use disorder unit, depending upon services provided. Refer to InterQual® Acute Pediatric Criteria or Refer to InterQual® Acute Adult Criteria

30:

Seizures are a sign of acute withdrawal from alcohol, sedatives, hypnotics, and anxiolytics (; American Psychiatric Association, Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition Text Revision, DSM-5-TR, 2022; Day and Daly, Addiction 2022, 117: 804-14; Wood et al., JAMA 2018, 320: 825-33).

31:

Seizures can be a sign for developing delirium tremens (Day and Daly, Addiction 2022, 117: 804-14; Grover and Ghosh, J Clin Exp Hepatol 2018, 8: 460-70; Wood et al., JAMA 2018, 320: 825-33). Inpatient detoxification is recommended for individuals presenting with alcohol related withdrawal complications such as seizures (National Institute for Health and Clinical Excellence (NICE), Alcohol-use disorders: diagnosis and clinical management of alcohol-related physical complications. Clinical guideline 100. 2010).

32:

Alcohol withdrawal syndrome and an oral temperature of at least 101°F (38.3C) may increase the risk for developing delirium tremens (Funayama et al., BMC Psychiatry 2021, 21: 387; American Society of Addiction Medicine, The ASAM Clinical Practice Guideline on Alcohol Withdrawal Management. 2020).

33:

Persistent vomiting refers to vomiting that is prolonged in duration and is still active at the time of evaluation, and can be a symptom of severe alcohol withdrawal (Day and Daly, Addiction 2022, 117: 804-14; American Society of Addiction Medicine, The ASAM Clinical Practice Guideline on Alcohol Withdrawal Management. 2020; Wood et al., JAMA 2018, 320: 825-33).

34:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

35:

Individuals with a CIWA-AR score of 15 or greater may be at increased risk for developing severe withdrawal (Attilia et al., Riv Psichiatri 2018, 53: 118-22). High CIWA scores are also associated with an increased risk of progression to a complicated withdrawal (American Society of Addiction Medicine, The ASAM Clinical Practice Guideline on Alcohol Withdrawal Management. 2020).

36:

A primary goal in the management of alcohol withdrawal syndrome (AWS) is the avoidance of further injury related to complications. Delirium tremens (DTs) is a serious complication of AWS with a mortality rate of 5-15% (0.05-0.15). Patient status and comorbidities help dictate the medication management of AWS. Benzodiazepines are the established choice for prevention and treatment of DTs. Antipsychotics may be used in conjunction with benzodiazepines to control psychiatric symptoms, such as anxiousness, hallucinations, and combativeness (Chandini, MJMSR 2013, 4: 86-9; DeBellis et al., J Intensive Care Med 2005, 20: 164-73).

37:

Moderate withdrawal symptoms include severe anxiety, and moderate to severe tremor. Severe or complicated withdrawal symptoms include seizure, delirium and hallucinations. Patients with severe and complicated withdrawal are appropriate for the critical level of care.

38:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO₂) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO₂. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

39:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or

work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

40:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

41:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula

Room air 21%(0.21)

1L 24%(0.24) 4L 36%(0.36)

2L 28%(0.28) 5L 40%(0.40)

3L 32%(0.32) 6L 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS

Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

42:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

43:

Refractory seizures, also known as uncontrolled, intractable, or drug-resistant seizures, are resistant to treatment and are poorly controlled with medication. Recurrent seizures are seizures that occur more than once and may occur without warning (Billington et al., J Clin Med 2016, 5:).

44:

Brand names for dexmedetomidine include:

- Precedex
- Dexdor
- Dexdomitor
- Sileo

45:

Continuous intravenous infusion of dexmedetomidine, which is an alpha 2 adrenergic agonist, is used in the critical care setting to treat symptoms of opioid, benzodiazepine, and alcohol withdrawal. Dexmedetomidine can also be used as an adjunct to sedation to reduce the use of anesthetic requirements. Clonidine, although rarely administered as a continuous infusion, is another alpha 2 adrenergic agonist used for similar purposes (J Addict Med 2020, 14: 1-72).

46:

Instruction: This criterion refers to the medically supervised medication treatment regimen that is used to purge the body of addictive substances. Protocols can include dosing schedules that are front-loading, fixed-dosing, or symptom-triggered. Medications can include:

- Benzodiazepines (e.g., chlorthalidopoxide, diazepam, lorazepam, and oxazepam)
- Barbiturates (e.g., phenobarbital)
- Sympatholytics (e.g., propranolol, clonidine, and atenolol)
- Anticonvulsants (e.g., carbamazepine, gabapentin, and divalproex)
- Other complementary medications used to manage the symptoms of withdrawal (e.g., antivirals such as amantadine)

47:

Evidence suggests certain anticonvulsants may be an effective treatment in preventing withdrawal progression, delirium, and seizures. Examples of anticonvulsants include, but are not limited to, carbamazepine, gabapentin, phenobarbital, and divalproex (American Society of Addiction Medicine, The ASAM Clinical Practice Guideline on Alcohol Withdrawal Management. 2020; Nisavic et al., Psychosomatics 2019, 60: 458-67).

48:

Chemical restraint is an emergency medication used to control behavior that presents an imminent threat of serious physical injury to self or others and is not part of the routine plan of care (Koukopoulos Alexia, Empathy, Normalization, and De-escalation 2021: 112). Routes of administration may include, but are not limited to, intramuscular or intravenous. Whenever a chemical restraint is utilized, it is important to identify the triggers and warning signs leading up to the dangerous behavior and to consider changes to the treatment plan to avoid continued emergency medication use.

49:

Patients with heavy alcohol use can develop a cross-tolerance to benzodiazepines. Barbiturates alone or in combination with benzodiazepines offer another alternative for the treatment of alcohol withdrawal syndrome

(AWS) (Wolf et al., Open Access Emerg Med 2020, 12: 53-65; Mo et al., J Crit Care 2016, 32: 101-7). An example of a barbiturate is phenobarbital.

50:

Examples of benzodiazepines include, but are not limited to, chlorthalidone, diazepam, lorazepam, and oxazepam.

51:

Opiate antagonists assist with alleviating opioid withdrawal symptoms with minimal sedation (Gowing et al., Cochrane Database Syst Rev 2017, 5: CD002021). Examples of opiate antagonists include, but are not limited to, naltrexone and naloxone. Extended-release naltrexone is being used for opioid withdrawal as well as relapse prevention (Bisaga et al., Am J Addict 2018, 27: 177-87).

52:

Instruction: Opioid antagonists are used for dual diagnosis withdrawal. Patients presenting solely with opioid withdrawal do not usually require acute medical treatment.

53:

Examples of opioid agonists include, but are not limited to, methadone, and partial opiate agonists such as buprenorphine, and butorphanol. Buprenorphine can also be a mixed agonist/antagonist when coupled with naloxone or naltrexone (American Society of Addiction Medicine, ASAM 2020 National Practice Guidelines for Treatment of Opioid Use Disorder. 2020; Srivastava et al., Lancet 2020, 395: 1938-48; Gowing et al., Cochrane Database Syst Rev 2017, 5: CD002021).

54:

Sympatholytics can be used to treat withdrawal symptoms from opioids (Kuszmaul et al., J Am Pharm Assoc 2020, 60: 145-52; Srivastava et al., Lancet 2020, 395: 1938-48). Examples of sympatholytics include, but are not limited to, propranolol, clonidine, lofexidine, and atenolol.

55:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

56:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

57:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

58:

A hallucination is a sensory perception that occurs without external stimulation of the sensory organ (e.g., vision, hearing, touch, taste, smell). Patients may or may not be able to distinguish between the hallucination and reality. Hallucinations can cause significant personal distress and can negatively impact daily functioning. Types of hallucinations include:

- Auditory: a false perception of sound, typically voices that are experienced as coming from either inside or outside of one's head and are different from one's own thoughts. Command hallucinations consist of a false perception of a voice telling the patient to do something, typically destructive to oneself or others
- Gustatory: a false perception of taste, usually unpleasant
- Olfactory: a false perception of smell, usually unpleasant
- Somatic: a false perception of a physical experience within the body
- Tactile: a false perception of being touched or of something on or under the skin (e.g., bugs crawling)
- Visual: falsely perceived images, either formed (e.g., people) or unformed (e.g., flashes of light)

References:(American Psychiatric Association, Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition Text Revision, DSM-5-TR, 2022; McCarthy-Jones et al., Front Psychol 2016, 7: 1418)

59:

Delusions are fixed beliefs that are not amenable to change in light of conflicting evidence. Delusion related distress and impairment in functioning may present in patients with schizophrenia, bipolar disorder, or obsessive compulsive disorder (Bortolotti, Delusions in Context. 2018).

60:

Impulsivity is defined as a lack of control or an inability to refrain from acting upon wishes, fantasies, thoughts, and feelings; impulsivity often results in negative consequences. Impulsivity may occur as a symptom while a patient is experiencing withdrawal from alcohol or substances (Reichl et al., Subst Use Misuse 2022, 57: 649-55; Nutt et al., Nutrients 2021, 13;; Kozak et al., Ann N Y Acad Sci 2019, 1451: 71-91).

61:

Psychomotor agitation (PMA) is excessive motor activity in association with an inner feeling of tension. The activity is usually repetitive and nonproductive (i.e., pacing, fidgeting, wringing of hands, inability to sit still). Evidence suggests PMA which is not managed in a timely manner may lead to harm to self or others (American Psychiatric Association, Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition Text Revision, DSM-5-TR, 2022; Ortiz-Sandoval et al., Int J Environ Res Public Health 2022, 19;; Pompili et al., Int J Environ Res Public Health 2021, 18;; Vieta et al., BMC Psychiatry 2017, 17: 328).

62:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

63:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

64:

Hepatic encephalopathy (HE) is a complication of acute or chronic liver disease that is primarily characterized by changes in personality, consciousness, cognition, and motor function. West Haven criteria for grading HE includes (Montrief et al., Am J Emerg Med 2019, 37: 329-37; Weissenborn, Drugs 2019, 79: 5-9; Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47):

- Grade I - Lack of awareness, anxiety, short attention span, euphoria, inability to add/subtract
- Grade II - Disorientation for time, inappropriate behavior, lethargy/apathy, obvious personality change, dyspraxia, or asterixis
- Grade III - Confusion, bizarre behavior, gross disorientation, responsive to verbal stimuli, somnolence to semi-stupor
- Grade IV - Coma

65:

While an elevated ammonia level is the classic laboratory finding used to diagnose hepatic encephalopathy, serial ammonia measurements are inferior to clinical assessment in gauging improvement or deterioration in a patient receiving treatment (Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47).

66:

Instruction: Patients with end-stage liver disease are generally on chronic outpatient lactulose, rifaximin, or neomycin at home to control symptoms of hepatic encephalopathy (HE). Admission to a skilled nursing facility (SNF) may be appropriate to initiate or adjust the dosage of these medications due to worsening hepatic encephalopathy.

67:

With Grade II hepatic encephalopathy, patients may experience disorientation for time to 3 of 5 of the following: day of the month, day of the week, month, season, and/or year.

68:

Oral branched-chain amino acid (BCAA) enriched formulations may improve the manifestations of occult or minimal episodic hepatic encephalopathy (HE). Intravenous administration is ineffective (Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47; J Hepatol 2014, 61: 642-59). Although BCAA has a beneficial effect on HE, there is no evidence regarding an effect on mortality, quality of life, or nutritional parameters (Gluud et al., Cochrane Database Syst Rev 2017, 5: CD001939).

69:

Lactulose is recommended by the American Association for the Study of Liver Diseases/European Association for the Study of the Liver guidelines as the first-line treatment for episodes of hepatic encephalopathy. It is a nonabsorbable substance used to treat an elevation in blood ammonia (NH_3) and reduces plasma ammonia concentration by converting soluble ammonia into insoluble ammonia in the colon. The laxative effect of lactulose results in the expulsion of trapped ammonium in the form of diarrhea. Lactulose can be administered orally, via a nasogastric tube, or as a retention enema (Swaminathan et al., Hepat Med 2018, 10: 1-11; J Hepatol 2014, 61: 642-59).

70:

Evidence has shown that L-ornithine L-aspartate (LOLA) is effective in lowering ammonia and improving mental status in patients with cirrhosis and hepatic encephalopathy (HE), with the oral form of LOLA being especially beneficial (Butterworth et al., J Clin Exp Hepatol 2018, 8: 301-13). In a 2017 randomized, controlled trial, patients with persistent HE showed improvement from intravenous LOLA administration, whereas oral supplementation was determined to be ineffective (Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47). A recent systematic review reassessed both oral and intravenous (IV) LOLA for the treatment of low-grade or minimal hepatic encephalopathy (MHE) and high grade or overt hepatic encephalopathy (OHE). The relative efficacy of LOLA depended on the type of HE as well as the route of administration. Patients with MHE, defined as patients with cirrhosis and alterations in mental status diagnosed by psychometric testing, appear to benefit more from the oral administration of LOLA than patients with high grade overt HE (Weissenborn, Drugs 2019, 79: 5-9; Butterworth et al., J Clin Exp Hepatol 2018, 8: 301-13).

71:

The renal and ototoxicity of neomycin have rendered this medication nearly obsolete in the treatment of hepatic encephalopathy (Tabbers et al., J Pediatr Gastroenterol Nutr 2014, 58: 258-74).

72:

Rifaximin is an anti-infective recommended by evidence-based clinical guidelines as an effective add-on therapy to lactulose for the treatment of overt hepatic encephalopathy (OHE). It has been shown to reduce healthcare resource utilization over the long term by lessening OHE recurrences and rehospitalization (Hudson and Schuchmann, European journal of gastroenterology & hepatology 2019, 31: 434). In a systematic review comparing the effectiveness and safety of interventions in patients with OHE, Rifaximin showed the greatest reduction in blood ammonia concentration (Zhu et al., Aliment Pharmacol Ther 2015, 41: 624-35).

73:

Instruction: Examples of antiarrhythmics with proarrhythmic risk include, but are not limited to, amiodarone, disopyramide, dofetilide, dronedarone, flecainide, mexiletine, propafenone, and sotalol. Many factors influence the appropriate choice of drug, including underlying medical conditions and concurrent drug administration. In some

instances, such as patients who are pregnant or are breastfeeding, there are drugs which are contraindicated or have limited evidence to support their use.

74:

West Haven criteria for hepatic encephalopathy (HE) includes the criterion "bizarre behavior" for Grade III HE. This behavior may be viewed as "out of sync" from societal norms; it varies from patient to patient and is often communicated to a healthcare provider by a patient's family member or caregiver (Ferenci et al., Hepatology 2002, 35: 716-21).

75:

Gross disorientation is defined as being disoriented to time and space (Biggins et al., Hepatology 2021, 74: 1014-48; Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47).

Disorientation to time includes at least 3 of the following:

- Disoriented to the date of the month
- Disoriented to the day of the week
- Unable to discern the month or season
- Disoriented to the year

Disorientation to space includes all of the following:

- Unable to identify the country
- Unable to identify the city, state, or region
- Disoriented to place

76:

Instruction: Patients with end-stage liver disease are generally on chronic outpatient lactulose, rifaximin, or neomycin at home to control symptoms of hepatic encephalopathy (HE). Hospitalization may be appropriate to initiate or adjust the dosage of these medications due to worsening HE.

77:

Refractory arrhythmias, also known as uncontrolled, intractable, or drug-resistant arrhythmias, are resistant to treatment and are poorly controlled with medication. Recurrent arrhythmias are arrhythmias that occur more than once and may occur without warning.

78:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

79:

Blood pressure is considered labile when it is extremely variable and difficult to control.

80:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

81:**Expected progress after 4 days**

- Withdrawal symptoms resolving and manageable with medications
- Vital signs stable or within acceptable limits
- Fluid, electrolyte imbalances, and nutrition returning to acceptable limits
- Labs returning to acceptable limits
- Dose and frequency of medications decreasing
- Adequate oral nutrition
- Activities at or close to baseline
- Social consult (e.g., medication/drug programs, social support services, emotional, legal, or vocational)

issues)

- Preparing for discharge within the next 24 hours

If no, consider

- Nutritional consult for patients with inadequate oral intake
- Changing route of medication from parenteral to PO
- Specialist consult (e.g., pancreatitis, diabetes mellitus, cirrhosis)
- Neuro consult and CT or MRI for persistent abnormal mental status

82:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

83:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

84:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

85:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

86:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

87:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

88:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

89:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

90:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

91:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house

extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

92:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

93:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

94:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

95:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

96:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)

- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

97:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

98:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

99:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

100:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

101:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

102:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784).

Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.